Review: Machines that Think by Inga Strümke 🇳🇴

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Abstract

A.C. Grayling describes humankind at a radical inflection point, facing a future that is a chaos of uncertainties. The response taken here is that no future is inevitable: Timothy Snyder has shown how a belief in inevitability collapses into its counterpoint, eternity — here the tyranny of technology — and that we retain both agency and a modicum of rationality. The question is not what the future holds but what we want our futures, plural, to look like, and whether we can reach a consensus on the values behind them.

Against that backdrop this is a review of Inga Strümke's *Machines that Think*, a guide through how AI developers think and what they must weigh. It takes up the fact that nobody knows precisely how models of billions of parameters arrive at their outputs, the explainable-AI research that treats the problem as a kind of forensics, and the consequences for jobs, mental health, Big Tech, mathematics and ethics.

Written by someone who worked from log books and slide rules to LLMs, and who now uses an AI agent daily while remaining wary of how seductive that convenience is.

> Humankind stands at a redical inflection point in its history. Almost everyone agrees that this is so, and unlike some widely held views, it is true. The future is a chaos of uncertainties. There are some who would go further and say that numanity stands not just at an inflection-point, but at its final point, at the verge of > > self-destruction.

With these words A.C. Grayling (Grayling (2026)) captures the apprehension and expectation of an AI, or more broadly, a technological future. What can we expect, is it possible to peek into the future? And if we don't like what we see, how much control should we, indeed, can we, exercise to determine an outcome that is worth living for?

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My own viewpoint is that no future is inevitable. Timothy Snyders has argued persuasively how inevitabilty leads to its destructive counterpoint, eternity - in this case the tyranny of technology. However frightening uncertain the future may seem, we have to "look worst-case scenarios squarely in the face, [otherwise] we will not prepare ourselves sufficiently to avoid or mitigate the worst that tomorrow can bring"[1]. Grayling offers comfort,

[1]See, for example the following quote,

> It starts at the bottom, and it keeps going up, … And I don't know where it stops.

> two things encourage us to think that we could survive the changes rushing at us, not least from the new technologies that, in the bewildering rapidity of their development, are currently making the future deeply, and in a number of ways threateningly, unreadable.
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> The first encouragment is that we are not without agency or a modicum of rationality, and could if we robust direct these unnervingly rapid changes in ways that reduce their dangers and leverage the benefits they also promise.
>
> The second is that to live without hope of survival is impossible; indeed, to hope for something better than mere survival is a necessity: hope is life, and life is the business of working to make hope come true.

Even if there is an existensial tension between hope an despair, we cannot abandon hope. We also cannot reset the clock, so to speak, return to a crossroads where we can try the other turn. Nor do we want to. Not everything from the past is worth preserving, not everything should be consigned to the rubbish bin. The question becomes, what do we want our futures (plural, there cannot be a single, universal ideal) to look like? How do we come to a broad concensus about our values and expectations, and how do we go about realising it?

Perhaps we operate behind John Rawls's "veil of ignorance", where none of us can be sure about our status in the new society, where our present expertise is no longer needed or appreciated. But every single person would want to have the best life possible. Us humans have at least a "modicum of rationality" and to abandon hope is to abandon life. It is in our power to preserve our agency, to resist surrendering our autonomy to the seductive power of modern technology.

Make no mistake, modern technology is powerful and powerfully addictive. Having lived from the times when the the calculations for physics experiments were done with log books and sliderules, when we had to connect through an operator ("number pleas!"), when bank reconds were analogue (your balance was entered by hand into the paper bank records with your personal bank book as backup), to pocket calculators (how proud we were of our HP calculators with its Reverse Polish Notation, how convenient!), to mainframes, the first PC's, the internet, smart phones, and LLM's. There is no question that the present AI agents make life in many respects a lot easier and companies more productive. Although one of last hold-outs, even I am using Claude on a daily basis to implement software solutions. This is where it becomes irresistably seductive: It is all too easy to ask Claude to implement a specific functionality, after which if will go away to think ("think"?!). After a few times asking for your approval (mechanically given, it might tell you that it now understands ("understands"?!) the problem well enough to come up with a solution that you will then, mechanically, approve. If you prompt your agent in the right way, it may even politely inform you that in the course of its own investigation it came up with the following inconsistencies or even straightout bugs. And, again very politely,

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— Umesh Ramakrishnan chief strategy officer at Kingley Gate

offer to fix it for you if you like. A far cry from log books and sliderules! And yes, I was there and has no desire in the least to go back to the bad old times.

What is there not to like about such a clever and polite assistant and how easy to be seduced into thinking that it knows everything and is infinitely wise (on top of being really, really clever), that mechanically accepting its suggestions will always turn out to be allright.

Most of what I do is quite straightforward and it takes only a little effort to go carefully through the code and check for correctness. Also, I try and work in areas where I understand the fundamental principles even if not alwys the precise details. In short, I'm working on problems with solutions that I know how to implement, with more or less effort. This helps me to guide the agent step-by-step through the process and makes it relatively straightforward to check the result. But I can also envisage situations where the problem becomes so complicated that the result will be hard, if not impossible to check and I'll be tempted to totally rely on the provided solution.

My whole career has been about understanding myself; I see no point in a situation where only the agent truly understands, or believes to understand, the problem. I can also easily envisage a situation where the effort of trying to understand will be too hard, too unproductive, too not worth the effort. Where we leave everything to the agent and spend our (forced) leisure time on social platforms. Where is the fun in that?

I hope I was successful in convincing you the reader that it is a jungle out there, one where anyone can get lost. No single person can be the ultimate guide out of the jungle. I would therefore suggest, join the debate, have as many informed discussions as possible, and do not loose sight that we have agency, that we have or should have ultimate control over our minds.

> … people stop relying on their own judgment. They surrender to whoever speaks with the most confidence, the most consistency, the most certainty. They choose the story that explains everything, because the alternative—that the world is complex and they have to think carefully—is too exhausting.
>
> — Hannah Arendt

Having that off my chest, you can now sigh with relief as I'm finally ready to start with the actual reason for this essay, namely to give you an inkling what *Machines that Think* is all about.

> I believe that basic knowledge about artificial intelligence will quickly become crucial for understanding much of what is happening in our lives and for being able to actively participate in the public debate about how we want this technology to influence our society.

— Inga Strümke

In short, Inga sets out to provide this basic knowledge so that you can join the debate.

0.a. Setting the Scene.

LLMs have become immensely powerful. So much so that people *say* 'AI' but *think* LLMs (Large Language Models. This is natural given how central communication is to the human condition. Humans depend on verbal, on written, and also the more abstract forms of communication such as poetry, music, painting, sculpture. LLMs (Large Language Models might only be a part of AI but it is a big part.

In just a few short years LLMs (Large Language Models have upended society. I want to point out just a few more recent developments, in no particular order. I'll return to them later on in the review. Let's start with the perhaps, the most obvious.

0.a.i. *Jobs:*

The debate is not so much about whether AI *is* disrupting the job market, as about which sectors are being hit the hardest and what the future holds for humans. Some industry leaders feel that humans may become redundant, others sound a more cautionary note. But there is no doubt that the landscape is changing[2], and even the experts don't always agree on where things are heading. There is no doubt that some sectors will be hit harder than others with junior software developers particularly vulnerable. How does one become a senior software developer without first being a junior developer[3]?

The idea is that AI will create more jobs than what it destroys, or that people will find more productive ways to spend their time[4].

Is this realistic, will people have more leisure time? By *leisure time* I don't mean *out of a job*, that is something else. It is more likely that expectations will also grow. Increased productivity means greater profit for you and your clients. If you happen to be in the consulting business your clients expect more for less and if your

[2] See, for example the following quote,

> It starts at the bottom, and it keeps going up, … And I don't know where it stops.
>
> — Umesh Ramakrishnan chief strategy officer at Kingley Gate

[3]

> According to one Stanford study, in the occupations most susceptible to greater automation, such as software engineering or customer service, entry-level employment has already declined by 16 percent relative to the least-exposed roles.
>
> — David M. Soloman, Chairman and Chief Executive of Goldman Sachs

[4]

> People are going to be more productive, they will have more time for leisure.
>
> — Sundar Pichai, CEO Google

competitors are willing to deliver, so must you. It is possible that you will work as hard as before but with less job satisfaction[5].

0.a.ii. *Mental Health:*

As far back as 2017 tech leaders expressed concern about the effect of technology on the human mind[6].

> We hardened our financial institutions against hacking … but it never occurred to us that the minds of our voters could be hacked; they turned our tools against us.
>
> — Roger McNamee, Managing Director of Elevation Partners

Brain hacking is of course nothing new. In *Influence: The Psychology of Persuasion*, for example, Cialdini (2007) describes how sales people hijack our natural and necessary psychological instincts to convince us to do stuff that we would not normally do. What is different now, is the scale and precision with which individuals are targeted. Sometimes to devastating effect, as the widescale interference in the 2016 US election testifies, see for example, Snyder (2018).

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[5]It is perhaps worth quoting the experience of Siddhant Khare at length,

> AI genuinely makes individual tasks faster. That's not a lie. What used to take me 3 hours now takes 45 minutes. Drafting a design doc, scaffolding a new service, writing test cases, researching an unfamiliar API. All faster.
>
> But my days got harder. Not easier. Harder.
>
> The reason is simple once you see it, but it took me months to figure out. When each task takes less time, you don't do fewer tasks. You do more tasks. Your capacity appears to expand, so the work expands to fill it. And then some. Your manager sees you shipping faster, so the expectations adjust. You see yourself shipping faster, so your own expectations adjust. The baseline moves.
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> Before AI, I might spend a full day on one design problem. I'd sketch on paper, think in the shower, go for a walk, come back with clarity. The pace was slow but the cognitive load was manageable. One problem. One day. Deep focus.
>
> Now? I might touch six different problems in a day. Each one "only takes an hour with AI." But context-switching between six problems is brutally expensive for the human brain. The AI doesn't get tired between problems. I do.
>
> — Siddhant Khare on 8 February, 2026

[6]

> [Tech] companies began with a goal of connecting the world. But once they put a business model in place that depended on advertising, they had to put in techniques that depended on creating addiction. Then when the smartphone came along, it created an opportunity to create a level of brain hacking that had never previously been seen.
>
> We hardened our financial institutions against hacking … but it never occurred to us that the minds of our voters could be hacked; they turned our tools against us.
>
> — Roger McNamee, Managing Director of the Venture Capital Firm Elevation Partners

Another study points out the ease with which humans are pursuaded by AI agents. Surely this is related to one of the psychological responses that Cialdini describes in his book, namely, the trust in authority and how that natural psychological reaction can be hijacked to ill effect - trust in authority can be misplaced.

A study conducted at Brown University indicates that chatbots systematically violate mentall health ethics standards.

It is worth emphasising the addiction to technology - tech companies still make most of their money from advertisements. This in turn, means that the user has to be kept engaged and happy. This is a theme that I'll return to later as it has consequences.

0.a.iii. *Big Tech:*

Big tech has become truly gigantic with no end in sight. The present economic growth in the US is largely driven by big tech and specifically AI companies. With their companies behind them, the tech leaders have become monstrously influential and ego-driven. Their decisions have planet-wide influence, and are not always, perhaps rarely, in the best human interest. We are starting to see push-back, from resistance to develop new data centers, to the booing of commencement speakers when they mention AI, to Pope LeoXIV's encyclical, Magnifica Humanitas. Perhaps one can add the EU AI Act of 2024and the General Data Protection Regulation, aiming to place limits on the use of AI and safeguarding the private data of individuals.

It is worth noting that the US is finding it extraordinally difficult to come up with similar legislation, and then there are autocratic countries where the public enjoy no protection whatsoever.

0.a.iv. *AI as a Mathematician:*

Quite recently OpenAI created quite a stir when on of their AI models solved an open mathematics problem that researchers have been looking at for many years. For some, this is an indications the dream of AGI is within reach, or has arrived. What makes it even more impressive is that a general LLM was used meaning, the model was not specifically trained for solving mathematical problems. It was a general-purpose model that just happened to be excellent at solving mathematical problems as well.

First a little background.

The Hungarian mathematician Erdős (1946) proposed the so-called unit distance conjecture, the technical details need not concern us here. Ever since mathematicians have tried to solve the conjecture without succcess[7]. That is, until recently when the conjecture was disproofed by an internal model at OpenAI. Most of the following discussion is based on Alon et al. (2026).

[7]

To start off, I have no inside view on the OpenAI process involved in the disproof. To put it in proper context, it is necessary to know, for example, how much and/or what type of human intervention was involved; was this the first problem given to the model, or was this the one from many that the model was able to solve, etc.

Despite this, it is possible to demystify the proof a bit, just using the information, taken from Alon et al. (2026).

Quite a few things came together on this problem that falls naturally inside the strengths of LLMs. Here I can do no better than quote Thomas Bloom

1. Although many first class mathematicians worked on the problem,

   most of the human efforts spent on this problem have been on trying to prove the upper bound, rather than spending serious time on trying to disprove it.

   On examining the construction, it becomes more clear how people had missed this before – it requires the confluence of several different unlikely events: that a good mathematician is (1) spending significant time in thinking about the unit distance conjecture in the first place; (2) seriously trying to disprove it, despite the oft-repeated belief of Erdős that it is true; (3) believes that there is mileage in generalising the original construction to other number fields, and so is willing to expend significant time in exploring such constructions; and (4) sufficiently familiar with the relevant parts of class field theory to recognise that the appropriately phrased question about infinite towers of number fields with appropriate parameters can be solved using existing theory.

   The AI met all of these criteria, and its success here echoes previous achievements: it often produces the most surprising results by persevering down paths that a human may have dismissed as not worth their time to explore, combining superhuman levels of patience with familiarity with a vast array of technical machinery. — Thomas Bloom

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AI agents are not indimidated by the reputation of great mathematicians,
and have a sufficiently wide overview that lacks most individual
researchers.
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2. It is noticeable that a team of experts could verify the solution, simplefy it and generalise it somewhat, in a relatively short time. There seems to be consensus that the team of this paper, have the necessary background to have solved the problem themselves, if they had the incentive to get together to work in it.
3. Many mathematicians work on developing new theory, as opposed to solving problems. There is also a tendency for silo-ing, as noted by Daniel Litt[8],

> … there are several reasons that explain why AI tools can be better than humans in finding such a construction. With or without a full agreement with these reasons, the fact is that the AI was able to do here what lots of excellent human researchers tried and failed to do. — Noga Alon

[8] In Alon et al. (2026).

> incentives towards specialization and silo-ing,though understandable, have cost us some high-quality science.
>
> In my own area—algebraic and arithmetic geometry—there are simply very few practitioners, and so arguably all problems are attention-bottlenecked. On the other hand, my sense is that it is unusual for well-known open problems to have answers accessible to clever argument, as opposed to new theory. Many of us work on programs, rather than problems.

— Daniel Litt

The remarks above do not mean to distract from the significance of the disproof of a major conjecture by an AI agent and the reader is encouraged to consult Alon et al. (2026) for more persepctive.

I have, however, a few questions of my own and a bit of speculation.

In this case it was comparatively straightforward to verify the AI result. If, as is the expectation, AI will develop super human powers and come up with results where human mathematicians find it very hard or impossible to verify it, what then? Will the results be accepted if, let's say, they were verified by another AI agent, keeping in mind that both were trained on essentially the same data? Apart from the pleasure and the aesthetic satisfaction mathematicians derive from mathematics it does not appear to me a game that we play. Rather, is an endeavour by humans for humans. In that sense it is very different from say, chess, where humans continue to compete agains one another, knowing full well that they don't stand a chance against modern algorithms. Mathematics is not a competitive game against other humans. To what extent are we willing to sacrifice human autonomy if, perhaps a big if, algorithms do become much stronger mathematicians than the best humans?

0.a.v. *Wrapping up:*

The point is, we all have a stake in the present and future developments of AI and a wide discussion of how humans can keep control, where we want to go, is undoubtedly called for.

> *As long as we avoid these discussions and fail to make deliberate decisions, these decisions will either be made by leaders of tech companies, or they won't be made at all. I don't know which is worse.*
>
> — Inga Strümke

> *…is has essentially become your civic duty to know something about AI*
>
> — Keith L. Downing

On the other hand, Jeremy Nixon who founded A.G.I. House (yes, the A.G.I. refers to Articicial General Intelligence) in 2022 with Andrey Karpahy, seems to be denying us

our "civic duty" in his dismissal[9] of the efforts of Pope LeoXIV's to draw attention to the wider impact of AI,

> It did not seem like the church had thought deeply about what their independent perspective was on A.I. It is not clear they could do that even if they tried.
>
> They couldn't have a position on it, because they don't understand it.
>
> — Jeremy Nixon

Let me repeat that, "They could not have a position on it because they don't understand it". We can have a full discussion of this quote elsewhere. For now, kindly note how easy it is to dismiss those that the tech leaders deem to be uninformed (perhaps not all tech leaders are equally knowledgable about disciplines outside their field. I can't help returning to the "silo-ed" effect mentioned by Daniel Litt above).

Having that off my chest, it is important to enter any discussion as informed as possible – the more informed the discussion the better the chances that something good will come from it.

0.b. Review

In her book, *Machines that Think* Inga takes you, the reader, on a fascinating journey through the weird and wonderful world of AI. She is not only pointing out the highlights but give you a peek behind the scenes. You will learn how the AI developers think about what they are doing, what are the issues any developer must considered when developing and applying AI models. As a highly respected, active researcher she is a reliable guide she is a reliable and enthusiastic guide through what is often a maze of conflicting opinions. She is enthusiastic about AI can offer, and has offered already, and is cautious when she must.

One of the things many people will find surprising, even a little frightening, is that no one knows exactly how these complex models with their billions if not trillions of parameters, 'think'. We know in principle how the models are trained but it is much harder to know what design choices the developers inside the tech companies make. Few of the models are open sourced that will allow one to inspect the details. But the deeper reason is that machines and humans operate on very different principles[10]. The models are not designed to be human interpretable. If we do not understand how they 'think' it is much harder to anticipate how they will perform in challenging environments. What is the risk of unknow consequences.

Fortunately for the rest of us there are researchers like Inga who is researching this exact problem. This research field, known as XAI can succintly be described as AI forensics. In this field, researchers dissect the internals of the large models to figure out what it is that they actually learn. As mentioned above, this is crucial to

[9] Not all tech leaders are dismissive, e.g. Christopher Olah, Benjamin Rossman.

[10] Of course, statements like these are controversial. Some, like Geoffrey Hinton argues that the models are already conscious. (What are the implications if we accept this?) For me, Neil Lawrence (Lawrence (2024)) makes a strong case for differences between machine and human intelligence.

understand where things can possibly go wrong, but is also crucial for learning from the models' learning.

Although a book of about 250 pages, it is rich in information and deep in scope. You will learn about the technicalities behind the models - but don't despair, you will be carefully guided with brillaint explanations - but also what it means for us as individuals, for societies and even the broader impact on humanity.

One of the constant themes throughout the book, is her emphasis of the role of *society* as a moderating influence on the future direction of AI. I'll discuss this fundamental issue in more detail, later. The emphasis on the importance of society and *civic duty* I referred to above, is, unfortunately, far from a universal value. Yet, the prevailing societal values determine to a large extent the path to the future, good or bad. AI therefore forces us to re-examine our basic values, as individuals, as societies

> Among the most fascinating aspects of articial intelligence as technology is how it forces us to contemplate difficult ethical dilemmas.
>
> — Inge Strümke.

Let's get into specifics.

0.b.i. *History:*

> **Note**
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> To be continued

0.c. A Deep Dive into Selected Topics.

0.c.i. *Ethics:*

Any discussion about the ethics of AI seems to start with the trolley problem. The variant used by Inga is the one of an unstoppable trolley running along a railway track straight at 5 people fastened. onto the track. You, as a bystander, happens to stand next to a control that will divert the trolley in time onto a different track. Sadly, on this track one person is tied down and will sure be killed if you activate the control. The question is, what is the moral behaviour in this situation, what will you do? For me, this problem is slightly awkward where there is no right answer. For me, I would like to have a third option and that is to try and stop the trolley, an option ruled out from the start. I find support from Rawls (1999) who rejects the unitarialism approach where one person is sacrificed for the greater good. Instead, he poses,

> Each person possesses an inviolability founded on justice that even the wellfare of society as a whole cannot override.
>
> — John Rawls

Applying this the trolley problem to the AI controlling self-driving cars, the question

becomes “Which life should be prioritised”[11]? My personal response to this question is, it is the moral duty of the AI scientist to ensure that this choice is not necessary by avoiding any collision at all cost, as I’m sure all of them do. As Inga points out, machines, in a critical situation do not face the same limitations as humans. They do not become stressed and have a far faster response time. This means that they should avoid the choice, which one to kill, the elderly person in the cross road, the baby, the passenger in the car?, until physics makes it impossible to avoid the collision. And if it is not possible to avoid the collision what are the guidelines for the scientist implementing the decision? This is when the scientist is faced with her own ethical and moral position. Ethics in AI is unavoidable.

We have already discussed the problems involved with the use of data. We have learned that the data should be representative of the population where the model is implemented. We have also seen how hard it is to foresee when a model my, without our awareness, introduce say, gender, bias.

I want to go in a different direction.

> **Note**
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> To be continued

0.c.ii. *A peek into the future:*

> **Note**
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> Incomplete

References

Alon, N., Bloom, T. F., Gowers, W. T., Litt, D., Sawin, W., Shankar, A., Tsimerman, J., Wang, V., & Wood, M. M. (2026,). *Remarks on the disproof of the unit distance conjecture*. https://arxiv.org/abs/2605.20695

Cialdini, R. B. (2007). *Influence: The Psychology of Persuation. Revised Edition*. Harper Collins.

Erdős, P. (1946). On sets of distances of n points. *American Mathematical Monthly, 53*, 248–250.

Grayling, A. (2026). *The Challenge of the Future. What Should we Keep from Yesterday as We Rush into Tomorrow?*. OneWorld Publications.

Lawrence, N. D. (2024). *The Atomic Human. Understanding Ourselves in the Age of AI*. Allen Lane.

Rawls, J. (1999). *A Theory of Justice*. Harvard University Press.

Snyder, T. (2018). *The Road to Unfreedom: Russia, Europe, America*. Penguin Random House.

[11] Machines that Think, p167.